

LESSON 114 (0.7 GROUND and 1.3 DUAL)

Lesson Objective

The purpose of this lesson is to introduce ground reference maneuvers and slips. This lesson will also enhance the student's division of attention and wind correction skills by introducing S-turns. Slips are practiced for use in emergency situations, in order to clear obstacles during the landing approach and crosswind situations.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - The student should be preparing for the mock stage 1 check, lesson 119.
 - Determine if the student has any specific areas that require further attention.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 07 - Ground Reference Maneuvers.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
- Warrior III Pilot's Information Manual.
 - Section 07 - Description and Operations of the Airplane and its Systems.

Lesson Content (Ground)

Operation of Systems - Avionics and Electrical

- Electrical System:
 - Alternator:
 - Main source of electrical power:
 - 14 volts, 70 amps (Warrior Cadet).
 - 28 volts, 70 amps (Warrior III).
 - Rotating electromagnet (rotor), connected to the engine crankshaft by alternator belt, generates electrical AC power as it spins past copper windings (stator).
 - Initially, a small electrical charge is provided by the battery to energize the electromagnet (field excitation current), then the alternator power is self-sufficient.
 - Field excitation current magnetizes the rotor initially.
 - The rotor turns inside the stator.

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- Electrical current is now produced from the alternator.
- Stand-by alternator (Warrior III - Avidyne):
 - Used as a backup to the alternator.
 - 28 volts, 20 amps.
- Battery:
 - Used as a backup to the alternator and for engine starting.
 - 12 volts, 35 amp (Warrior Cadet).
 - 24 volts, 35 amps (Warrior III).
 - Recharged by the alternator.
 - Located at the right forward face of the firewall, inside a thermoplastic box.
 - Charging rate:
 - Normal: 2 A.
 - After start-up: much higher.
 - The 35-amp hour battery can theoretically provide 35 amps for 1 hour.
 - Based on a new battery and electrical system.
 - Expect half.
- Ammeter:
 - Displays load in amperes placed on the generating system.
 - With only the battery master switch on, the ammeter will show the amount of charging current demanded by the battery.
 - It never shows the discharge rate of the battery.
- Voltage Regulator:
 - Located on the forward left side of the fuselage, behind the instrument panel.
 - Controls alternator output.
 - Keeps voltage between 11-14 volts (Warrior Cadet).
 - Keeps the voltage to a max of 31 volts in the Warrior III.
 - Controls by varying the current sent to rotor (electromagnet):
 - More current results in a stronger magnetic field which increases alternator output.
 - Less current results in a weaker magnetic field which decreases alternator output.
- Overvoltage Relay:
 - Located on the forward left side of the fuselage, behind the instrument panel.
 - The relay opens and takes the alternator offline to protect the electrical system.
 - 17 volts (Warrior Cadet).
 - 31 volts (Warrior III).
- Circuit breakers.
 - The number on the front of the circuit breaker indicates how many amps can flow through it at a maximum.

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- If the amperage exceeds that number, then the breaker will disconnect and not allow electricity to flow through it.
 - The circuit breakers should never be reset more than three times in a flight. This is to avoid an electrical fire.
- 1 bus bar.
- External power:
 - Can be plugged in to outside of the aircraft if the battery is unable to provide enough power for the starting process.
- Avionics (Introduce the ones specific to the airplane that the student is in):
 - Garmin 430:
 - Requires 24 volts, not 12.
 - 2 power converters convert power from the 12-volt primary system to 24 volt to power the system – called step up transformer.
 - Provides all-in-one GPS/Nav/Comm.
 - Garmin G5:
 - Internal Components:
 - Air Data Attitude Heading Reference System (ADAHRS):
 - Accelerometers.
 - Magnetometer.
 - Avidyne Entegra:
 - Internal components - Primary Flight Display (PFD):
 - Air Data Computer (ADC):
 - Uses pitot static system and OAT information for indicated airspeed, true airspeed, altitude, and vertical speed.
 - Transducers sense air pressure and convert it to digital information.
 - Attitude Heading Reference System (AHRS):
 - Uses an accelerometer and magnetometer for attitude, rate of turn, and heading information.
 - Magnetometer is located in the left wing.
 - Takes about 3 minutes to align itself.
 - If reset in flight, after 30 seconds AHRS will not align until stationary.
 - Internal components - Multi-Function Display (MFD):
 - Data Acquisition Unit (DAU):
 - Receives and displays engine information digitally.

S-Turns

- The student should follow along with the Airplane Flying Handbook chapter 6.
- Ground reference maneuver in which the airplane's ground track forms two opposite and equal half-circles on each side of a ground-based straight-line reference.
- Select a prominent straight-line ground-based reference point (perpendicular to the wind).
 - Mention obstacle avoidance and landing point in case of emergency.

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- Enter on the downwind at 1,000 feet.
- Use smoke, water, PFD, groundspeed, etc. to determine where the winds are coming from.
- Emphasize space and time to set up!
- throughout the maneuver.
- Objectives of the maneuver:
 - Adjusting the bank angle accordingly in correlation with the groundspeed.
 - Maintaining a specific distance between the airplane and the reference point.
 - Division of attention.
 - Compensating for wind drift.
- Common errors:
 - Failure to clear the area.
 - Failure to enter the maneuver at a constant altitude and airspeed.
 - Failure to maintain altitude during the maneuver.
 - Failure to assess wind direction.
 - Failure to maintain a constant radius turn.
 - Failure to maintain airplane coordination.

Slips: Forward/Side

- Forward:
 - A forward slip is performed to reduce altitude more rapidly without gaining airspeed.
 - This is accomplished using the ailerons to lower the upwind wing while simultaneously using the rudder in the opposite direction.
 - Keeping the nose aimed at the aiming point should be emphasized to the student, as a nose high attitude can result in a cross-controlled stall.
 - Power must be at idle.
 - Anticipate the recovery from the slip in order to resume normal glide path.
- Side:
 - Used to maintain a desired ground track in conditions with wind present.
 - This is accomplished using the ailerons to lower the upwind wing while simultaneously using the rudder in the opposite direction to keep the aircraft lined up with centerline.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices,

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life jackets, fly away kit, survival kit, etc.)

- FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal/Crosswind Takeoff

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

Review Previous Exercises as Required

- Review any areas the student might need help with by utilizing the pertinent lesson plans.

S-turns

- First completion of the maneuver will be a demonstration.
 - Instructor does, student says.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
- Second completion of the maneuver will be performed by the student
 - Student says, student does.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
 - Provide feedback as needed throughout the maneuver.

Slips: Forward/Side

- Forward:

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- Student says, student does.
 - Verify correct procedure.
 - Guard against overbanking or using too much rudder.
- Side:
 - Student says, student does.
 - Verify correct procedure.
 - The airplane must be aligned with the centerline.
 - Guard against overbanking or using too much rudder.

Normal Traffic Pattern

- The student should be able to establish and execute a proper traffic pattern with minimal instructor guidance.
- It will likely still be necessary for the instructor to provide guidance during the round out and flare portions.

Normal/Crosswind landing

- The student should conduct the landing.
- Provide guidance during the round and flare.
- Provide guidance on proper crosswind correction.
 - Emphasize that this must be maintained even after touchdown.

Go-around/Rejected Landing

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 115.
- Assign homework:
 - Warrior Maneuver's Manual.
 - Review non-towered operations.
 - Review steep turns.
 - Review and memorize the emergency checklists.
 - Give a copy of the completed V speed and Emergency quiz for student to learn/memorize

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Perform ground reference maneuvers with instructor guidance.
4. Conduct forward and sideslips with instructor guidance.
5. Conduct traffic patterns, takeoffs and landings with little instructor guidance.